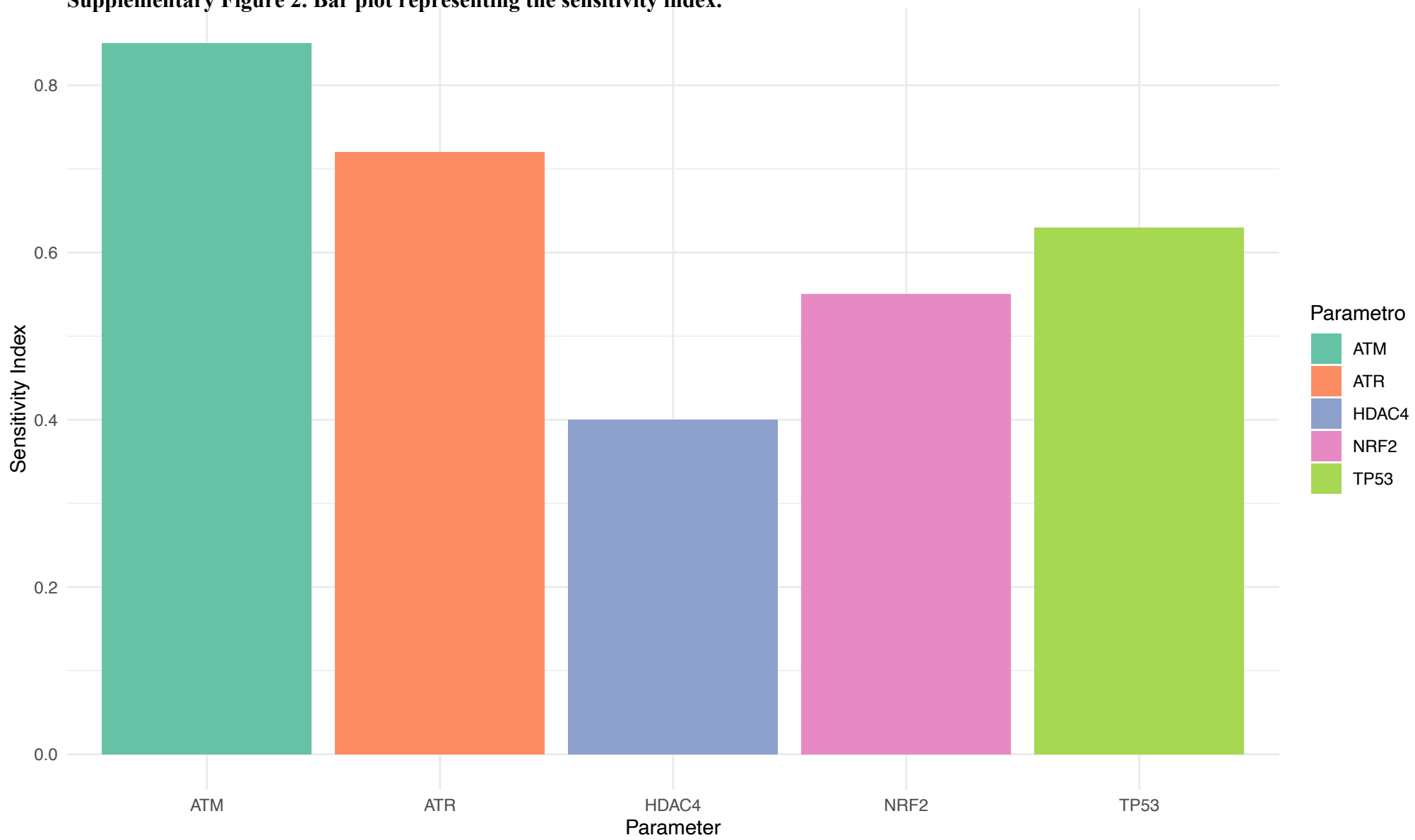


Supplementary Figure 2. Bar plot representing the sensitivity index.



Parameter	Value	Reaction / Role	Kinetics Type	Biological Activity	Biological Rationale/ Justifications	Parameter Source
k_activation	0.01 l ² /(mol·s)	ATM activation	Mass-action	Models cooperative activation of ATM dimers	ATM signaling is detectable ~5 minutes post-damage . In modeling terms, a similar order-of-magnitude activation rate was used by Fedak, Elizabeth A et al., who assumed a maximal ATM activation of ~0.5 per minute (~0.0083 s ⁻¹). Our value is of the same order, reflecting a reasonable estimate to achieve rapid ATM activation consistent with experiments.	1, 2
k_deactivation	0.1 l/s	ATM, ATR, CHK deactivation	First-order	Represents constitutive dephosphorylation or degradation	The chosen rate is an intentional simplification to ensure these kinase signals are transient. While faster than biological half-lives, it effectively represents robust phosphatase activity and protein turnover that terminates ATM/ATR/CHK1	3, 4

					signaling.	
k_repair	0.05 l ² /(mol·s)	DNA repair by ATM	Mass- action	Simplified representation of repair enzyme action	This moderate rate was chosen so that when ATM is active, the “DNA damage” species is cleared on a timescale of minutes. This is qualitatively consistent with experimental observations that a large fraction of DSBs is repaired within tens of minutes to a couple of hours after damage when ATM is available	5
k_damage_production	0.1 mol/s	DNA damage production	Zeroth- order	Constant damage induction (e.g., oxidative stress)	The magnitude 0.1 (in model concentration units per second) is justified by known cellular damage rates: it's estimated human cells incur ~10 ⁴ spontaneous DNA lesions per day , which is on the order of 0.1 events per second per cell. Loft and Poulsen (1996) reported oxidative DNA modifications accumulating at ≈10 ⁴ per cell per day . Our	6

					value yields a similar scale of continuous damage input. It essentially assumes a low but steady induction of DNA damage (e.g. from oxidative stress) to the system, consistent with background damage in vivo.	
k_p53_activation	0.02 l ² /(mol·s)	p53 activation by ATM	Mass-action	ATM phosphorylates and stabilizes p53	We assign a second-order rate for this ATM-p53 interaction. This value is comparable to those used in previous p53 network models (typically ~0.01–0.02 for kinase→substrate phosphorylation when exact data are lacking) and produces the expected behavior that active ATM rapidly elevates p53 levels.	7
k_p53_deactivation	0.1 l/s	p53 degradation	First-order	p53 degradation via MDM2 pathway	Upon DNA damage, ATM/ATR phosphorylation extends p53 half-life from “minutes to hours” by preventing MDM2 binding. Our model	8, 9

					compresses these dynamics by using 0.1 s^{-1} ($t_{1/2} \sim 7 \text{ s}$) to ensure that active p53 decays quickly once ATM/ATR input is removed. This high rate effectively simulates the potent negative feedback by MDM2. The model's faster rate is a deliberate simplification to allow p53 levels to oscillate or drop on short timescales, matching the pulse-like dynamics seen in experiments.	
k_HDAC4_activation	$0.02 \text{ l}^2/(\text{mol}\cdot\text{s})$	HDAC4 activation	Mass-action	HDAC4 regulates gene expression in DDR	This mass-action rate is like other enzyme activation steps. This nominal value was tuned to allow HDAC4 to respond on a minute-timescale. (In general, when specific rates are unknown, modelers use educated guesses in the $0.01\text{--}0.1 \text{ M}^{-1}\text{s}^{-1}$ range; our choice falls in that range.)	2, 10

k_HDAC4_deactivation	0.05 l ² /(mol·s)	HDAC4 deactivation	Mass- action	Enzymatic feedback or phosphatase activity	This value was set slightly higher than the activation rate to ensure HDAC4 activity is transient. It is an order-of-magnitude estimate chosen for model stability.	11
k_HDAC4_p53	0.03 l ² /(mol·s)	p53 inhibition by HDAC4	Mass- action	HDAC4 deacetylates p53, reducing its activity	This value lies in a typical range for protein-protein enzymatic interactions. It ensures that when HDAC4 is active, it can noticeably decrease p53 activity. (Notably, HDAC inhibitors that block class I/II HDACs lead to increased acetyl-p53, confirming that HDAC mediated p53 deacetylation is a significant regulatory mechanism. Thus, including HDAC4→p53 with a non-negligible rate is biologically reasonable, even if exact kinetics are unknown.)	10, 12
k_I2PP2A_production	0.005 l/s	I2PP2A synthesis	Zeroth- order	Regulatory inhibitor synthesis	This value was likely chosen to yield a low	2, 13

					steady-state level of I2PP2A when balanced with its degradation. For context, many cellular proteins are produced at rates on the order of 10^{-3} – 10^{-2} (arbitrary units) per second in modeling studies to represent basal gene expression. Our value falls in that range.	
k_I2PP2A_degradation	0.01 l/s	I2PP2A degradation	First-order	Constitutive turnover	We set production and degradation such that I2PP2A reaches a steady-state around 0.5 “units” (since $0.005/0.01 = 0.5$). The rationale is to have a dynamic pool of I2PP2A that can respond to changes (e.g. get sequestered by binding). This is a reasonable generic value for protein turnover in modeling. It ensures I2PP2A levels can change over time rather than being constant.	2
k_binding	0.1 l ² /(mol·s)	PP2A and I2PP2A binding	Mass-action	Protein-protein binding	This value is an educated guess to represent a	14

					limited protein-protein binding affinity, ensuring that PP2A is not completely sequestered by I2PP2A at baseline.	
k_unbinding	0.01 l/s	Complex dissociation	First-order	Represents dissociation rate	This value is tuned so that the inhibition of PP2A by I2PP2A is significant but readily reversible, aligning with the idea that PP2A activity can be restored on a short timescale.	-
k_TOPBP1_activation	0.02 l ² /(mol·s)	TOPBP1 activation	Mass-action	TOPBP1 as ATR co-activator	This value is similar to ATM's own activation rate, so that when the upstream signal is present, TOPBP1 becomes active reasonably quickly. The exact value is a modeling assumption to integrate known qualitative behavior: that TopBP1 is rapidly activated in response to DSBs to turn on ATR.	2
k_TOPBP1_deactivation	0.1 l/s	TOPBP1 inactivation	First-order	Constitutive turnover	0.1L/s ensures TopBP1's active state lasts around	

					7 seconds, preventing sustained ATR hyperactivation. This value was chosen by analogy to ATM/ATR dephosphorylation rates, reflecting the rapid reversal of TopBP1's activation.	
k_ATR_activation	$0.02 \text{ l}^3/(\text{mol}^2 \cdot \text{s})$	ATR activation	Higher-order	Cooperative activation with TOPBP1	$0.02 \text{ L}^3/(\text{mol}^2 \cdot \text{s})$ reflects ATR's cooperative activation, requiring two ATR molecules and TopBP1 to form a complex. This rate, set to match ATM activation timescales, ensures ATR is only activated under specific conditions consistent with ATR's more complex and limited activation in biology.	1, 2
k_ATR_deactivation	0.1 l/s	ATR inactivation	First-order	Passive decay	We use the same 0.1 s^{-1} as for ATM, effectively assuming ATR is deactivated on the order of seconds in the model. It ensures that in	1, 15

					simulations ATR does not remain active without ongoing DNA damage.	
k_p53_activation_ATR	0.015 l ² /(mol·s)	ATR-mediated p53 activation	Mass-action	ATR pathway activates p53 via phosphorylation	We assign a second-order rate of 0.015 M ⁻¹ s ⁻¹ , slightly lower than ATM's rate (0.02) to reflect ATR's somewhat less prominent role in p53 regulation. It is chosen to allow ATR, when fully active, to stabilize p53, consistent with observations that ATR can activate p53 in the absence of ATM.	7
k_repair_ATR	5e-07 l ² /(mol·s)	ATR-mediated DNA repair	Mass-action	Very low efficiency reflects ATR's indirect repair role	By setting this rate so low, the model acknowledges ATR's role in DNA repair is "indirect and low-efficiency", matching qualitative expectations. The chosen value simply ensures ATR alone cannot rapidly clear DNA damage in the simulation.	5
k_CK2	0.02 l ² /(mol·s)	CK2-mediated HDAC4 phosphorylation	Mass-action	Post-translational regulation by CK2	0.02 was chosen to give CK2 significant regulatory impact	11

					HDAC4 without making the reaction instantaneous – consistent with a post-translational modification event.	
k_dephos	0.05 1/s	HDAC4 dephosphorylation	First-order	PP2A-like phosphatase effect	This rate, with a half-life of ~14 seconds, ensures HDAC4's phosphorylation status flips quickly, reflecting the fast action of phosphatases in reversing kinase activity. The value is tuned to prevent prolonged phosphorylation of HDAC4 after CK2 activity drops and is consistent with typical deactivation rates (around 0.1 s^{-1}), offering a reasonable, generic rate for dephosphorylation.	
alpha_spermidine	0.5 l/mol	Spermidine effect on CK2-TOPBP1	Modulatory constant	Modulates CK2 activation in presence of spermidine	Zhang et al. (2019) found that spermidine increases CK2 activity in a dose-dependent manner, with $4\text{ }\mu\text{M}$ boosting activity in cells.	16

					We model this using a modulatory factor $\alpha=0.5\text{L/mol}$, which enhances CK2 activity by 50% per unit of spermidine. This factor reflects experimental observations that millimolar spermidine can roughly double CK2 activity.	
k_CK2_TOPBP1	0.02 l/s	CK2-TOPBP1 mediated activation	First-order	Simplified catalytic model	This value is an estimate chosen so that CK2's presence significantly speeds up ATR activation via TopBP1. The numerical choice (0.02) parallels other activation rates in the model, indicating consistency in assuming moderate, on-the-order-of 10^{-2} s^{-1} kinetics for unspecified signaling steps.	-
k_CHK1_activation	$0.02 \text{ l}^2/(\text{mol}\cdot\text{s})$	CHK1 activation	Mass-action	Downstream ATR effector kinase	We model ATR-mediated CHK1 phosphorylation as reaction with $k=0.02 \text{ l}^2/(\text{mol}\cdot\text{s})$ Similarr to the rates for $\text{ATM} \rightarrow \text{p53}$ or $\text{ATM} \rightarrow \text{ATR}$. This rate ensures a	1

					moderate phosphorylation of CHK1 by ATR, with activation building up over tens of minutes, consistent with experimental observations.	
k_CHK1_deactivation	0.1 l/s	CHK1 inactivation	First-order	Natural decay	=0.1L/s was chosen to reflect the rapid inactivation of phospho-CHK1 through both phosphatases and proteasomal degradation. This rate ensures transient CHK1 activity, aligning with the biological reality that CHK1 signals are typically brief to allow the cell cycle to resume.	4
k_CHK2_activation	0.02 l ² /(mol·s)	CHK2 activation	Mass-action	ATM-dependent checkpoint kinase	Experiments show that CHK2 phosphorylation occurs within minutes after irradiation: “both ATM and CHK2 are activated as early as 10 min after irradiation, with ATM activity peaking before CHK2”. This fast timescale justifies a relatively high	17

					second-order rate constant (0.02 l ² /(mol·s)) so that CHK2 becomes active quickly once ATM is active	
k_CHK2_deactivation	0.1 l/s	CHK2 inactivation	First-order	Natural decay	0.1L/s was chosen to reflect the moderate deactivation of active CHK2, which can persist for hours unless actively dephosphorylated. Experimental evidence shows CHK2 activity remains detectable up to 16 hours post-damage, indicating sustained activation.	17
k_p53_activation_CHK2	0.02 l ² /(mol·s)	CHK2-mediated p53 activation	Mass-action	Reinforces p53 stabilization	This second-order rate was set similar to ATM→CHK2, assuming CHK2 rapidly phosphorylates p53 once active. The chosen value reflects the observed timescale of p53 activation (tens of minutes to a few hours), consistent with experimental findings that CHK2	17

					phosphorylation of p53 occurs early in the DNA damage response.	
k_NRF2_activation	$0.01 \text{ l}^2/(\text{mol}\cdot\text{s})$	NRF2 activation	Mass-action	NRF2 release from KEAP1 under stress	$0.01 \text{ l}^2/(\text{mol}\cdot\text{s})$ was chosen to model NRF2 activation, reflecting its release from Keap1 under oxidative or electrophilic stress. In vivo studies show that NRF2 activation can occur within an hour, with peak nuclear translocation observed around 1 hour after stress induction. The chosen rate ensures NRF2 activation is slower than direct kinase cascades like CHK2/p53, but still rapid, achieving substantial NRF2 release within 1–2 hours of a strong stress signal.	18, 19
k_NRF2_deactivation	0.05 l/s	NRF2 deactivation	First-order	Return to basal NRF2 levels	Our value 0.05 l/s (interpreted as $\sim 0.05 \text{ min}^{-1}$) corresponds to a $t_{1/2}$ in the order of minutes ($\sim 13 \text{ min}$ if 0.05 min^{-1}), in line with experimental half-lives	20, 21

					for NRF2 (10–30 min). This ensures that once the activating stress is gone, NRF2 activity falls rapidly (allowing the cell to reset to basal antioxidant levels within 1–2 hours). In essence, the parameter reflects Keap1-driven degradation rate of free NRF2.	
k_NRF2_repair	0.02 $\text{l}^2/(\text{mol}\cdot\text{s})$	NRF2-mediated DNA repair	Mass-action	Indirect activation of antioxidant repair genes	We assign $0.02 \text{ l}^2/(\text{mol}\cdot\text{s})$ to this mass-action term, paralleling other secondary activation rates, to reflect that NRF2's induction of repair processes is not instantaneous but occurs over time as target genes (HO-1, GCLC, 53BP1, etc.) are upregulated. The rationale is that this rate is tuned so that NRF2's benefit to DNA repair manifests on the scale of a few hours. This aligns with observations that activating NRF2 before radiation reduces DNA damage and	22

					chromosomal aberrations in cells, implying a measurable repair-enhancing effect within one cell cycle.	
k_NRF2_KEAP1	0.01 l ² /(mol·s)	NRF2 degradation via KEAP1	Mass-action	Cytoplasmic sequestration & degradation	We use a relatively large association rate (0.01 l ² /(mol·s) in our units) to ensure free NRF2 is rapidly captured by Keap1, keeping basal NRF2 activity low. This choice is supported by NRF2's short basal half-life and the floodgate model: only when Keap1 pools are saturated free NRF2 escape degradation.	23
K_omav	1 mol/l	Omaveloxolone EC ₅₀	Michaelis constant	Drug modulation of NRF2 pathway	Although omaveloxolone has an experimental EC ₅₀ reported in the low nanomolar range, our choice of 1 M is a modeling convenience – effectively assuming the drug is at saturating levels relative to this high K (so its effect can be toggled on). In this way, the drug effect term is mostly in a linear regime	24

					for any realistic concentration used, simplifying the model. In the absence of an exact in-model EC ₅₀ , we note that the qualitative behavior is that omaveloxolone at micromolar to nanomolar concentrations will significantly activate NRF2, as observed experimentally. Thus, the model's K _{omav} = 1 mol/L serves to scale the drug's influence without limiting it, given the much lower real EC ₅₀ values.	
k_spermidine_activation	0.05 l ² /(mol·s)	Autophagy activation by spermidine	Mass- action	Nutrient-sensing activation	This value reflects a relatively fast mass-action induction of autophagy by spermidine. In practical terms, this means that when spermidine is present (modeled as a species or input), autophagy activity rises rapidly (within an hour or so) – consistent with	25, 26

					observations that autophagy markers (LC3-II, etc.) increase shortly after spermidine treatment. This parameter is chosen so that even a moderate spermidine level can substantially activate autophagy, aligning with its role as a potent caloric-restriction mimetic that prolongs lifespan via autophagy.	
k_autophagy_deactivation	0.05 1/s	Autophagy decay	First-order	Constitutive degradation	Our choice of 0.05 1/s (if interpreted as $\sim 0.05 \text{ min}^{-1}$) gives a half-life of ~ 14 min for autophagy activity to decay, which is quite fast. This captures the idea that autophagy is dynamically regulated and will not remain high without sustained stimuli. Biologically, this can correspond to constitutive turnover of autophagy proteins/complexes and the action of phosphatases and mTOR signaling that rapidly	

					dismantle the autophagic machinery once stress/nutrient signals change. In summary, a value of 0.05 ensures autophagy activity falls to baseline on the order of minutes to an hour after an inducing signal is removed, consistent with observations of prompt autophagy termination upon refeeding.	
k_autophagy_repair	0.001 l/s	Autophagy-mediated DNA repair	Saturable	Catalytic repair rate	We model autophagy's effect on DNA repair as a saturable process (hence using Michaelis-Menten kinetics with k_autophagy_repair and K_m_autophagy). We set k_autophagy_repair = 0.001 l/ s, a relatively small rate, to reflect that this is a supportive, slower repair mechanism compared to dedicated DNA repair enzymes. Autophagy's impact on DNA lesion removal is indirect and gradual (e.g. mitigating	27

					oxidative damage, recycling components, or affecting HR vs NHEJ balance). No exact kinetic value is available, but we justify this choice by noting that autophagy's pro-repair benefits, while real, manifest over longer timescales. For instance, loss of autophagy causes genome instability that is evident over many hours or multiple cell cycles.	
Km_autophagy	10 mol/l	Autophagy Michaelis constant	Michaelis-Menten	Saturation threshold	In the absence of empirical K _m , we intentionally set a large K _m value so that under physiological ranges of damage or autophagy signal, the process operates far from saturation (i.e. first-order in the damage level). In other words, this K _m value ensures that autophagy's repair capacity is effectively unsaturated under normal conditions, allowing for a near-linear	

					relationship between autophagy activity and repair rate.	
k_p21	0.09 l/s	p21 production by p53	First-order	Cell cycle arrest mediator	We use a first-order rate of 0.09 l/s for p53-driven p21 production. In effect, when p53 is stabilized, p21 levels rise with a rate of ~ 0.09 (units s^{-1} or min^{-1} depending on time scale). If we assume minutes, 0.09 min^{-1} gives a half-time of ~ 7.7 min for p21 mRNA/protein accumulation towards a new steady state. That might be a bit fast for translation but note that this parameter also subsumes p21 mRNA processing and protein stabilization by p53. In many cell types, p21 protein is detectable by $\sim 4\text{--}6$h after DNA damage. Our chosen value yields significant p21 induction within a few hours, aligning with those observations.	28

k_p21_deg	0.01 l/s	p21 degradation	First-order	Protein turnover	We use 0.01 l/s as the first-order degradation rate, which (if taken as $\sim 0.01 \text{ min}^{-1}$) corresponds to a $t_{1/2} \sim 69 \text{ min}$ – within the reported range. This means that even as p53 drives p21 up, p21 levels will drop relatively quickly once p53 activity falls off. The value aligns with experimental data showing that p21 is largely gone a few hours after p53 levels decline, and that protein synthesis inhibitors reveal p21's short half-life ($\sim 30\text{--}45 \text{ min}$, which can extend to $\sim 2 \text{ h}$ if stabilized by certain interactions). Thus, $k_{p21_deg} = 0.01$ ensures transient p21 accumulation, consistent with the quick turnover observed <i>in vivo</i> .	29 30
k_BAX	0.05 l/s	BAX production by p53	First-order	Pro-apoptotic gene expression	$k_{BAX} = 0.05 \text{ l/s}$ is slightly lower than p21's rate but still a significant first-order rate. This	31

					yields BAX protein accumulation on the timescale of a few to several hours once p53 is active, accordingly with the notion that BAX protein levels start to rise during prolonged or high p53 activity. BAX induction is a bit slower or less immediate than cell-cycle arrest signals. Nonetheless, the value is high enough that a strong p53 signal (such as irreparable damage) will produce substantial BAX within one cell cycle to initiate apoptosis. Supporting this, BAX transcript upregulation is often observed within 6–12 h of genotoxic stress in experiments.	
k_BAX_deg	0.01 l/s	BAX degradation	First-order	Apoptosis resolution	We choose k_BAX_deg = 0.01 l/s. It implies that BAX levels will remain stable for multiple hours after induction, supporting the apoptotic program unless actively inhibited. This	32

					parameter is set low (slow decay) to reflect BAX's stability compared to p21. Reported BAX half-life in healthy cells is on the order of 8–12 hours, while in stressed or cancer cells BAX can be actively degraded, with half-life of 3-6h. Our attempt is to balance these two conditions.	
k_PUMA	0.03 l/s	PUMA production by p53	First-order	Apoptosis promoter	We assign $k_{\text{PUMA}} = 0.03$ l/s, a first-order rate for p53-driven PUMA production. This is in the same order as BAX's rate (slightly lower in our table), reflecting that PUMA induction is significant but might require strong p53 signals. Some studies suggest PUMA is induced as early as 4–6 h after p53 activation (earlier than Bax), consistent with PUMA's role as a potent activator of the pro-apoptotic Bax/Bak. Our chosen value produces an appreciable	32, 33

					PUMA level within a few hours of sustained p53 presence. Since PUMA is strictly p53-dependent in many contexts, having a non-zero k_{PUMA} emphasizes that, even if p53 pulses, PUMA can accumulate over multiple pulses to a threshold that triggers cell death. The value is a compromise – not as high as p21's (since PUMA might require higher p53 levels or more sustained stress), but high enough to ensure PUMA can reach functional levels on the timescale of a day or less, fitting experimental induction profiles.	
$k_{\text{PUMA_deg}}$	0.01 l/s	PUMA degradation	First-order	Turnover of apoptotic signal	We set $k_{\text{PUMA_deg}} = 0.01$ l/s, analogous to p21 and BAX in our table, which gives PUMA a ~1–2 h half-life, consistent with its reported stability. Thus, this parameter ensures PUMA levels drop relatively quickly	34

					when p53 signaling abates, preventing unnecessary apoptosis if damage is repaired. It also means that cells continuously require p53 activity to keep PUMA levels high; if p53 is only transiently on, PUMA will be produced but also degraded on the order of hours.	
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References

1. Zhao H, Li J, You Z, Lindsay HD, Yan S. Distinct regulation of ATM signaling by DNA single-strand breaks and APE1. *Nat Commun.* 2024;15(1):6517. Published 2024 Aug 7. doi:10.1038/s41467-024-50836-6
2. Fedak EA, Adler FR, Abegglen LM, Schiffman JD. ATM and ATR Activation Through Crosstalk Between DNA Damage Response Pathways. *Bull Math Biol.* 2021;83(4):38. Published 2021 Mar 11. doi:10.1007/s11538-021-00868-6
3. Shreeram S, Demidov ON, Hee WK, et al. Wip1 phosphatase modulates ATM-dependent signaling pathways. *Mol Cell.* 2006;23(5):757-764. doi:10.1016/j.molcel.2006.07.010
4. Zhang YW, Otterness DM, Chiang GG, et al. Genotoxic stress targets human Chk1 for degradation by the ubiquitin-proteasome pathway. *Mol Cell.* 2005;19(5):607-618. doi:10.1016/j.molcel.2005.07.019

5. Sun T, Li X, Shen P. Modeling amplified p53 responses under DNA-PK inhibition in DNA damage response. *Oncotarget*. 2017;8(10):17105-17114. doi:10.18632/oncotarget.15062
6. Loft S, Poulsen HE. Cancer risk and oxidative DNA damage in man [published correction appears in *J Mol Med* 1997 Jan;75(1):67-8]. *J Mol Med (Berl)*. 1996;74(6):297-312. doi:10.1007/BF00207507
7. Khosravi R, Maya R, Gottlieb T, Oren M, Shiloh Y, Shkedy D. Rapid ATM-dependent phosphorylation of MDM2 precedes p53 accumulation in response to DNA damage. *Proc Natl Acad Sci U S A*. 1999;96(26):14973-14977. doi:10.1073/pnas.96.26.14973
8. Gupta A, Shah K, Oza MJ, Behl T. Reactivation of p53 gene by MDM2 inhibitors: A novel therapy for cancer treatment. *Biomed Pharmacother*. 2019;109:484-492. doi:10.1016/j.biopha.2018.10.155
9. Moll UM, Petrenko O. The MDM2-p53 interaction. *Mol Cancer Res*. 2003;1(14):1001-1008.
10. Basile V, Mantovani R, Imbriano C. DNA damage promotes histone deacetylase 4 nuclear localization and repression of G2/M promoters, via p53 C-terminal lysines. *J Biol Chem*. 2006;281(4):2347-2357. doi:10.1074/jbc.M507712200
11. Intracellular Signaling to Histone Deacetylases - Cayman Chemical <https://www.caymanchem.com/news/intracellular-signaling-to-histone-deacetylases>
12. Roy S, Packman K, Jeffrey R, Tenniswood M. Histone deacetylase inhibitors differentially stabilize acetylated p53 and induce cell cycle arrest or apoptosis in prostate cancer cells. *Cell Death Differ*. 2005;12(5):482-491. doi:10.1038/sj.cdd.4401581
13. Wei Y, Maximov V, Morrissy SA, Taylor MD, Pallas DC, Kenney AM. p53 Function Is Compromised by Inhibitor 2 of Phosphatase 2A in Sonic Hedgehog Medulloblastoma. *Mol Cancer Res*. 2019;17(1):186-198. doi:10.1158/1541-7786.MCR-18-0485
14. Alsallaq R, Zhou HX. Prediction of protein-protein association rates from a transition-state theory. *Structure*. 2007;15(2):215-224. doi:10.1016/j.str.2007.01.005
15. Choi BK, Fujiwara K, Dayaram T, et al. WIP1 dephosphorylation of p27^{Kip1} Serine 140 destabilizes p27^{Kip1} and reverses anti-proliferative effects of ATM phosphorylation. *Cell Cycle*. 2020;19(4):479-491. doi:10.1080/15384101.2020.1717025

16. Ao Y, Zhang J, Liu Z, et al. Lamin A buffers CK2 kinase activity to modulate aging in a progeria mouse model. *Sci Adv.* 2019;5(3):eaav5078. Published 2019 Mar 20. doi:10.1126/sciadv.aav5078
17. García-Santisteban I, Llopis A, Krenning L, et al. Sustained CHK2 activity, but not ATM activity, is critical to maintain a G1 arrest after DNA damage in untransformed cells. *BMC Biol.* 2021;19(1):35. Published 2021 Feb 19. doi:10.1186/s12915-021-00965-x
18. Goldring CE, Kitteringham NR, Elsby R, et al. Activation of hepatic Nrf2 in vivo by acetaminophen in CD-1 mice. *Hepatology.* 2004;39(5):1267-1276. doi:10.1002/hep.20183
19. Ulasov AV, Rosenkranz AA, Georgiev GP, Sobolev AS. Nrf2/Keap1/ARE signaling: Towards specific regulation. *Life Sci.* 2022;291:120111. doi:10.1016/j.lfs.2021.120111
20. Ngo V, Duennwald ML. Nrf2 and Oxidative Stress: A General Overview of Mechanisms and Implications in Human Disease. *Antioxidants (Basel).* 2022;11(12):2345. Published 2022 Nov 27. doi:10.3390/antiox11122345
21. Nguyen T, Sherratt PJ, Huang HC, Yang CS, Pickett CB. Increased protein stability as a mechanism that enhances Nrf2-mediated transcriptional activation of the antioxidant response element. Degradation of Nrf2 by the 26 S proteasome. *J Biol Chem.* 2003;278(7):4536-4541. doi:10.1074/jbc.M207293200
22. Kim SB, Pandita RK, Eskiocak U, et al. Targeting of Nrf2 induces DNA damage signaling and protects colonic epithelial cells from ionizing radiation. *Proc Natl Acad Sci U S A.* 2012;109(43):E2949-E2955. doi:10.1073/pnas.1207718109
23. Liu S, Pi J, Zhang Q. Mathematical modeling reveals quantitative properties of KEAP1-NRF2 signaling. *Redox Biol.* 2021;47:102139. doi:10.1016/j.redox.2021.102139
24. Reisman SA, Gahir SS, Lee CY, Proksch JW, Sakamoto M, Ward KW. Pharmacokinetics and pharmacodynamics of the novel Nrf2 activator omaveloxolone in primates. *Drug Des Devel Ther.* 2019;13:1259-1270. doi:10.2147/DDDT.S193889
25. Madeo F, Bauer MA, Carmona-Gutierrez D, Kroemer G. Spermidine: a physiological autophagy inducer acting as an anti-aging vitamin in humans?. *Autophagy.* 2019;15(1):165-168. doi:10.1080/15548627.2018.1530929

26. Morselli E, Mariño G, Bennetzen MV, et al. Spermidine and resveratrol induce autophagy by distinct pathways converging on the acetylproteome. *J Cell Biol.* 2011;192(4):615-629. doi:10.1083/jcb.201008167
27. Gillespie DA, Ryan KM. Autophagy is critically required for DNA repair by homologous recombination. *Mol Cell Oncol.* 2015;3(1):e1030538. Published 2015 Oct 2. doi:10.1080/23723556.2015.1030538
28. Wu L, Levine AJ. Differential regulation of the p21/WAF-1 and mdm2 genes after high-dose UV irradiation: p53-dependent and p53-independent regulation of the mdm2 gene. *Mol Med.* 1997;3(7):441-451.
29. Chang LJ, Eastman A. Differential regulation of p21 (waf1) protein half-life by DNA damage and Nutlin-3 in p53 wild-type tumors and its therapeutic implications. *Cancer Biol Ther.* 2012;13(11):1047-1057. doi:10.4161/cbt.21047.
30. Abbas T, Dutta A. p21 in cancer: intricate networks and multiple activities. *Nat Rev Cancer.* 2009;9(6):400-414. doi:10.1038/nrc2657
31. Nakano K, Vousden KH. PUMA, a novel proapoptotic gene, is induced by p53. *Mol Cell.* 2001;7(3):683-694. doi:10.1016/s1097-2765(01)00214-3
32. Xin M, Deng X. Nicotine inactivation of the proapoptotic function of Bax through phosphorylation. *J Biol Chem.* 2005;280(11):10781-10789. doi:10.1074/jbc.M500084200
33. Yu J, Zhang L. PUMA, a potent killer with or without p53. *Oncogene.* 2008;27 Suppl 1(Suppl 1):S71-S83. doi:10.1038/onc.2009.45
34. Xie W, Zhang L, Jiao H, et al. Chaperone-mediated autophagy prevents apoptosis by degrading BBC3/PUMA. *Autophagy.* 2015;11(9):1623-1635. doi:10.1080/15548627.2015.1075688

Supplementary Table 2. Sensitivity and stability analysis protocol

Analysis	Key Settings (Problem & Method)	Output Report
<i>Time-Course Sensitivity Analysis</i>	Problem: Duration = 1 s; StepSize = 0.01 s; StepNumber = 100; list of parameter CN identifiers (e.g., ModelValue_0 = k_activation) in ListOfParameters; list of species CN identifiers (e.g., Metabolite_4 = DNA_damage) in ListOfTargets. Method: LSODA with sensitivities enabled; Relative Tolerance = 1×10^{-6}; Absolute Tolerance = 1×10^{-12}; Max Internal Steps = 10000.	Tab-delimited file containing time points and $\partial(\text{target})/\partial(\text{parameter})$ values at each integration step.
<i>Steady-State Sensitivity Analysis</i>	Problem: SubtaskType = 1 (steady-state); TargetFunctions = list of species (e.g., Metabolite_28 = p21, Metabolite_29 = BAX); ListOfVariables = list of parameter CN identifiers (e.g., ModelValue_39 = k_p53_activation_ATR). Method: Finite-difference Sensitivities Method; Delta factor = 0.001; Delta minimum = 1×10^{-6}; tolerances inherited from LSODA ($\approx 1 \times 10^{-6}, 1 \times 10^{-12}$); Max Internal Steps = 10000.	Tab-delimited file listing each parameter, each target species, and the sensitivity $\partial(\text{target})/\partial(\text{parameter})$ at steady state.
<i>Lyapunov Exponent Analysis</i>	Problem: ExponentNumber = 3; DivergenceRequested = true; TransientTime = 0 s. Method: Wolf's algorithm; Orthonormalization Interval = 1 s; Overall Integration Time = 1000 s; Relative Tolerance = 1×10^{-6}; Absolute Tolerance = 1×10^{-12}; Max Internal Steps = 10000.	Tab-delimited file reporting the first three Lyapunov exponents ($\lambda_1, \lambda_2, \lambda_3$) and divergence values if requested.

Supplementary Note 1. Supplementary reproducibility information.

- **Model file:** AT_ATM_ATR_forDR_v1.cps
- **Software:** COPASI version 4.44 (build 295)

(A) Time-Course Sensitivity Analysis

- Task: *Time Course*
- Duration: 1000 seconds; intervals: 100
- Sensitivity: Enabled
- Perturbed parameters: k_{ATM} , k_{ATR} , k_{p53} , etc.
- Output: Time-series CSV exported and visualized in R (Supplementary Figures 1 and 2)

(B) Steady-State Sensitivity Analysis

- Task: *Steady-State*
- Sensitivity: Enabled
- Global method: Central Difference
- Perturbation size: $1e-4$
- Output: Sensitivity matrix exported as `.txt` and visualized in heatmap (Figure 6)

(C) Lyapunov Exponents (Wolf Method)

- Task: *Lyapunov Exponents*
- Method: Wolf
- Number of exponents: 3
- Time: 1000
- Tolerance: $1e-6$
- Result: 3 negative exponents + average divergence reported in manuscript

Other specific details about the protocol are reported in Supplementary Table 2.

